Roll No. ....

TPE–013

B. TECH. (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

SATELLITE COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the following with respect to Satellite : 10 Marks (CO1)

(i) Satellite Axis

(ii) Ascending Node

(iii) Descending Node

(iv) Perigee

(v) Apogee

OR

(b) The orbit of an earth orbiting satellite has an eccentricity of 0.15 and a semi-major axis of 9000 km. Given that, the radius of earth 6371 km and μ is 3.986×10^{14} m^3/s^2, determine : 10 Marks (CO1)

(i) The Periodic Time

(ii) The Apogee Height

(iii) The Perigee Height

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2. (a) Explain the Kepler's law of planetary motion. Briefly discuss the various types of orbits. 10 Marks (CO1)

OR

(b) The sum of apogee and perigee distances of a certain elliptical satellite orbit is 50000 km and the difference of the apogee and perigee distances is 30000 km. Determine the eccentricity. 10 Marks (CO1)

3. (a) What do you understand by satellite subsystems ? Explain the Altitude and Orbit Control (AOC) subsystem. 10 Marks (CO2)

OR

(b) A satellite is moving in a circular orbit at a height of 200 km above the surface of earth. Determine its orbital velocity. Given that $G = 6.67 \times 10^{-11}$ Nm^2/kg^2, $m = 5.98 \times 10^{24}$ kg and radius of the earth is 6370 km. 10 Marks (CO2)

4. (a) Describe the telemetry, tracking and command (TTC&M) subsystems of a satellite system. 10 Marks (CO2)

OR

(b) A geostationary satellite is located at a distance of 3000 km with an operating frequency 14.25 GHz. The gain of transmitting and receiving antennas are 15 and 20 simultaneously. If the transmitter power is 200 kW, calculate the power received by the receiving antenna.

10 Marks (CO2)

5. (a) What is meant by look angle ? Explain them with reference to a geostationary satellite and earth station. 10 Marks (CO1 and CO2)

OR

(b) Determine the limits of visibility for an earth station situated at mean sea level at latitude 48.42° North and longitude 89.26° West. Assume a minimum angle of elevation for a geostationary satellite is 5°, radius of earth 6371 km. 10 Marks (CO1 and CO2)

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